

California Housing: Model Comparison

Task 2 model report

Objective

Predict median owner-occupied house value using eight California block-group features from scikit-learn's dataset.

Method

20,640 rows; 80/20 train-test split (seed 42)

Median imputation and StandardScaler fitted on training data

Linear Regression, Ridge (alpha=1), Decision Tree (depth=5)

Selected by lowest RMSE on 4,128 held-out rows

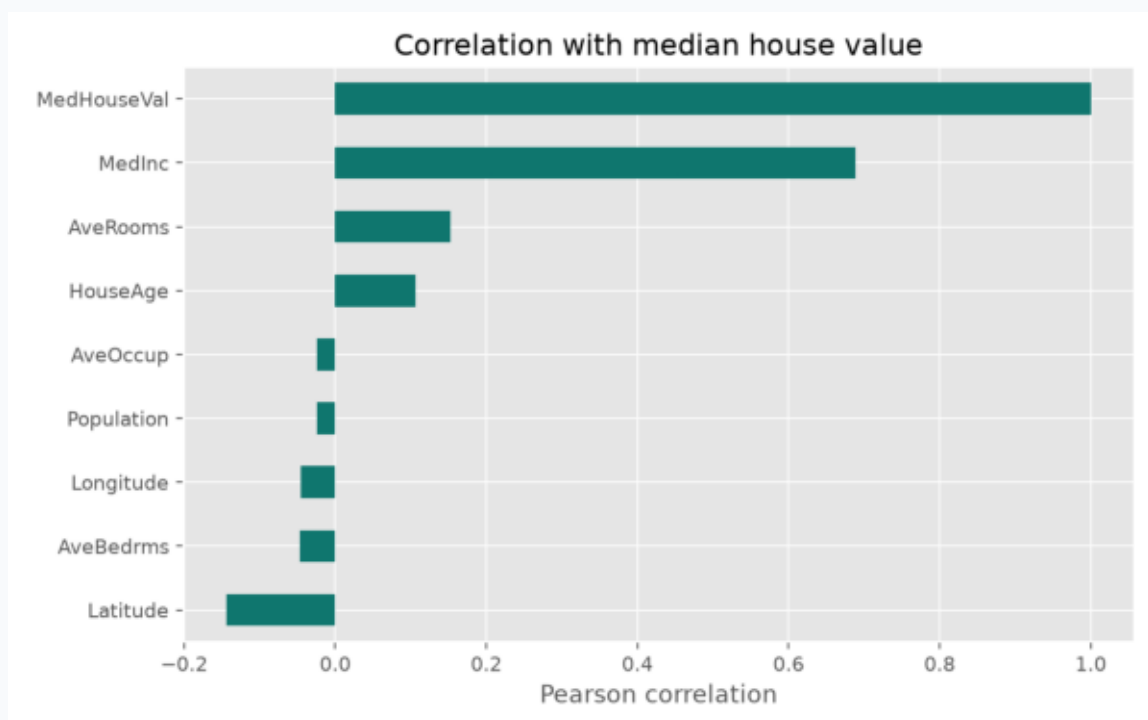
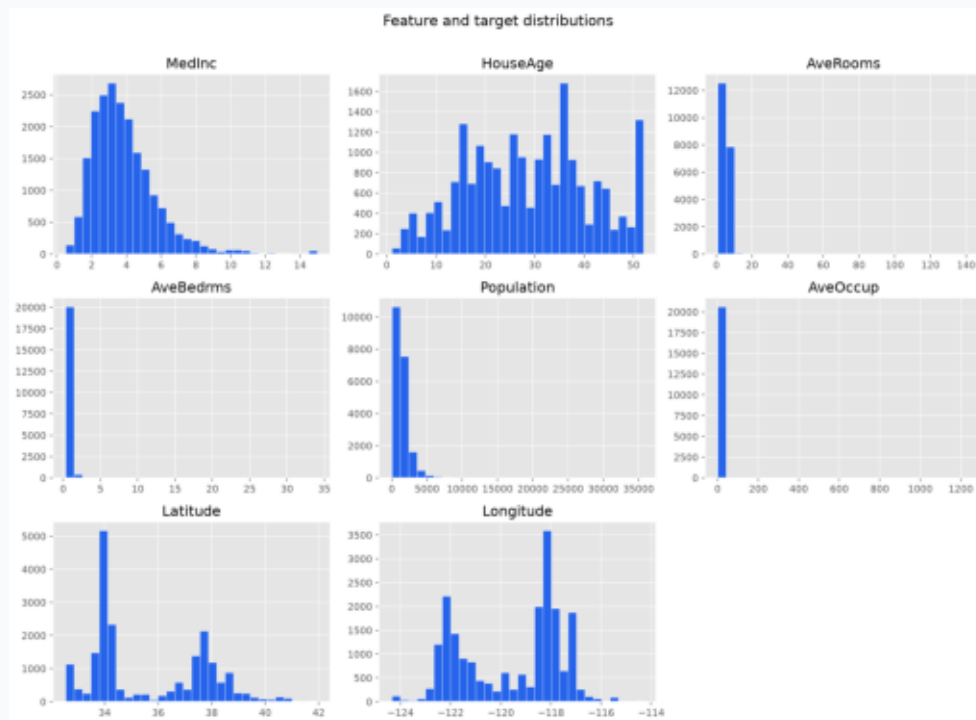
Selected: Decision Tree

MAE 0.5223 100k RMSE 0.7242 100k R-squared 0.5997

Target values are in \$100,000s of 1990 USD. This is an educational baseline, not a valuation tool.

Exploratory data analysis

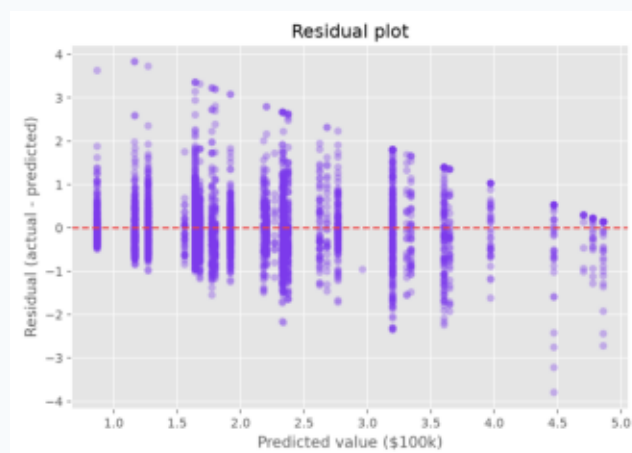
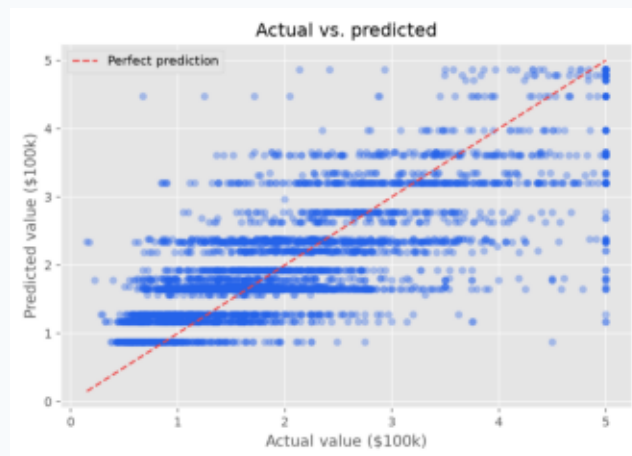
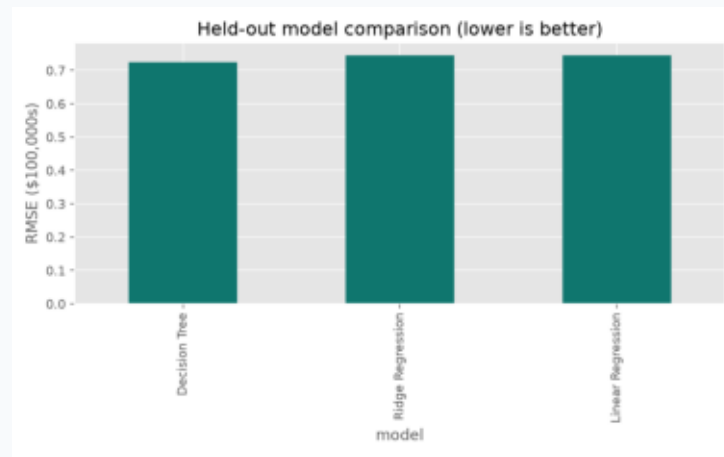
Distributions and feature relationships



Median income and location-related features show useful association with the target; the target's cap makes

Evaluation and model selection

Held-out comparison and prediction diagnostics



Selection uses one test split. Compare with cross-validation and newer property-level data before practical use